→ strings → holds the data of only 'i' type.

iterable. String with multiple words.

→ print words in python 2, 3

without print only in python 3

→ len ('Hello world')

↓

'11'

→ String indexing ✓

→ string slicing

↓

: → to perform slicing.

len(word).

s[: : 2]

↓ ↓ ↓

start end step size.

s[::] , s(:) → everything.

s[-3] → accessing from end.

s[:-1] , s[::-1] → reverse the string.

↓ ↓

start what is the difference

at beginning

and fetch till last

No slicing in interviews

String properties :-

↓

immutable object (can't change the value).

↓

can't modify data, not possible for item assignment.

concatination is possible.

(is a new reference created)??

S = S + 'x')'

↓

reassigning the string.

z = letter

letter * 3 = zzz

→ Built in string :-

S. upper (), S. lower (), S. split ()

↓

white space.

S. split (|) → any delimiter.

↓

delimiter is not included in the output.

S. join () → ?!

- format (), f"

→ location and counting.

S. count ('o'), S.find ('o')

→ expand tabs

\t ⇒ replaced with tab space.

→ is check methods.

S. isalnum ()

↳ checks for alpha or numeric.

S. isalpha ()

S. isspace ()

S. istitle ()

checks for first letter is upper.

S. endswith ('o')

↳ check for end.

→ reg exp.

S. partition () → similar to split but includes the delimiter.

5. split(.)

5. partition (.).

→ Lists

↳ different types of data types.

↳ mutable.

↳ editable

→ indexing, slicing → trace.

→ append, pop (list)

→ Nested list

↓

matrix

2-dimensional array

matrix [x] [y].

matrix [0] [0] [1]

↔ list comprehensions

first_col = [i [0] for i in matrix]

↓ ↓

first index of each list. each list in matrix

→ practice list comprehensions

→ Advanced list

l.count()

l.extend() (vs) append() : at the end

↓ combines the list

↓ list & list

insert()

↓ at specific position

remove()

↓ at specific position

(vs)

pop()

↓ at last

reverse(), sort():

↓ asc, desc

→ Tuples → permanent storage
→ Convert to list and edit
↓
immutable

→ sets (eliminate the duplicates).
↓
discard, remove → only if exists.
↓
pop remove

→ Clear, complete removal.

→ dictionary → hash tables.
↓
key value to pairs.
↓
handles duplication
↓
only takes the last assignments
ignores the initial assignments
↓
nested dictionaries.

dict [key] = value → accessing. → important.

d. keys
d. values
d. items

dictionary comprehensions

→ Shallow copy

import copy.

original.

shallow = copy.copy(original)

copy.deepcopy(original).